

Types of Data Structures: Linear & Non-Linear

Q. Classify Data Structures into Linear and Non-Linear types. Differentiate between them and provide examples of each.

****Definition to Remember****

A ****Data**
arrange
hierarch

1. Linear Data Structures

* Elements are arranged in ****sequential order**** - each element has a unique predecessor and successor.

* Can

Examples:

| DS | Description |

| :--- | :-

2. Non-Linear Data Structures

* Elements are arranged in a ****hierarchical or network**** structure.

* One

Examples:

| DS | Description |

| :--- | :-

3. Linear vs Non-Linear Comparison

| Feature | Linear | Non-Linear |

| :--- | :-

LINEAR: [A] -> [B] -> [C] -> [D]

NON-LINEAR (Tree):

[A]

*****Must-Write Points (for 10 marks)*****

1. Data

*****Quick Recall*****

`Linear
-> Tree,